

Mathematics – I (M-I)
GTU # 3110014



Dedicated Faculty, Committed Education

Darshan
Institute of Engineering & Technology

Unit – 1.3

Application of Definite Integral



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Sr. No.	Content	Total Hrs	% Weightage
01	Indeterminate Forms and L'Hôpital's Rule.	01	15 %
	Improper Integrals, Convergence and divergence of the integrals, Beta and Gamma functions and their properties.	03	
	Applications of definite integral, Volume using cross-sections, Length of plane curves, Areas of Surfaces of Revolution	03	

Unit 1.3 Applications of definite integral

- ✓ Volume using cross-sections(3 methods)
- ✓ Length of plane curves
- ✓ Areas of Surfaces of Revolution



Outline

- 
- ➡ Introduction
 - ➡ Slicing Method
 - ➡ Disk Method
 - ➡ Washer Method
 - ➡ Length of Plane Curves
 - ➡ Area of Surface by Revolution



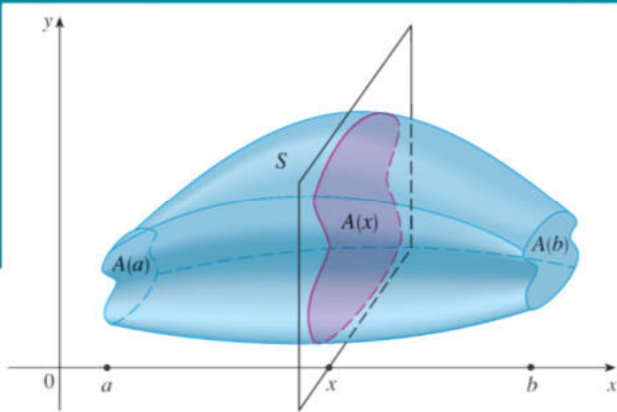


Introduction

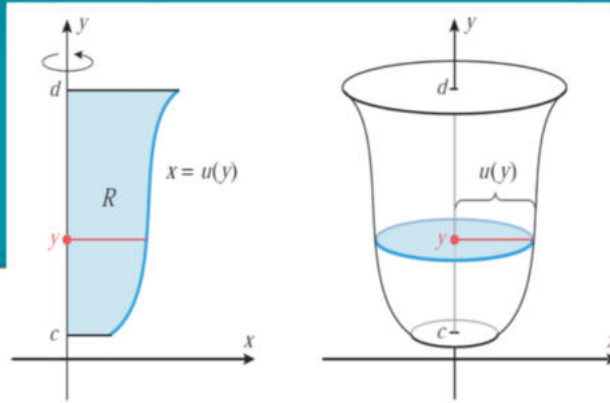


Volumes Using Cross-Sections

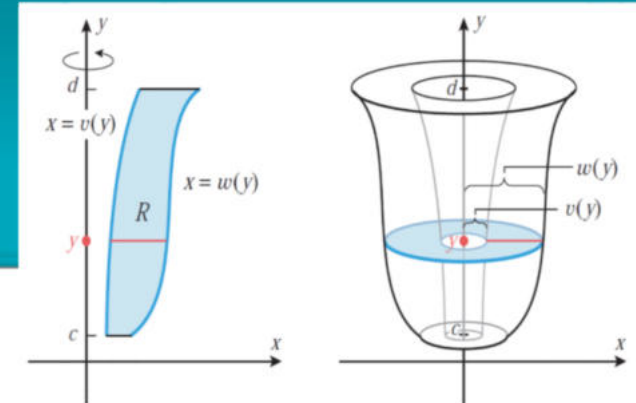
- ▶ In this section we define volumes of solids using the areas of their cross-sections.
- ▶ We learn three different methods to finding the volume of a particular solid.



The slicing method



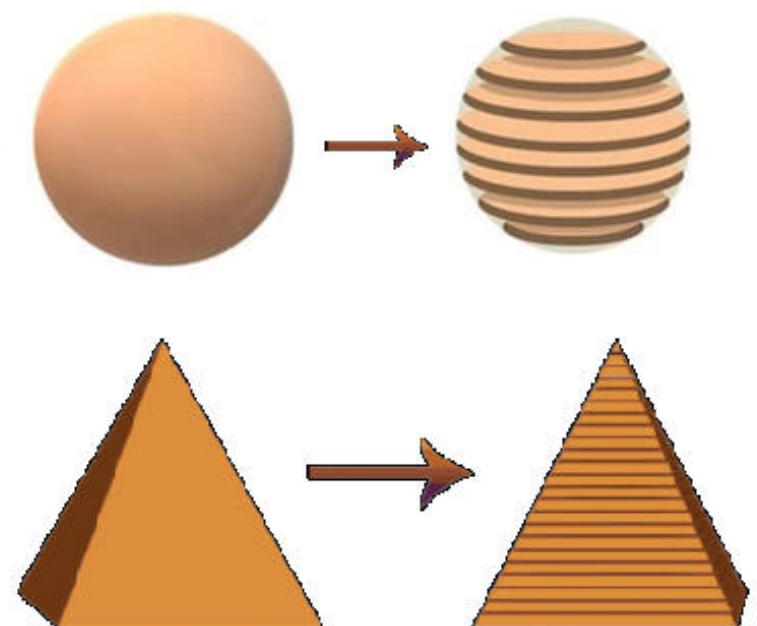
The disk method



The washer method

Volume of Solid by

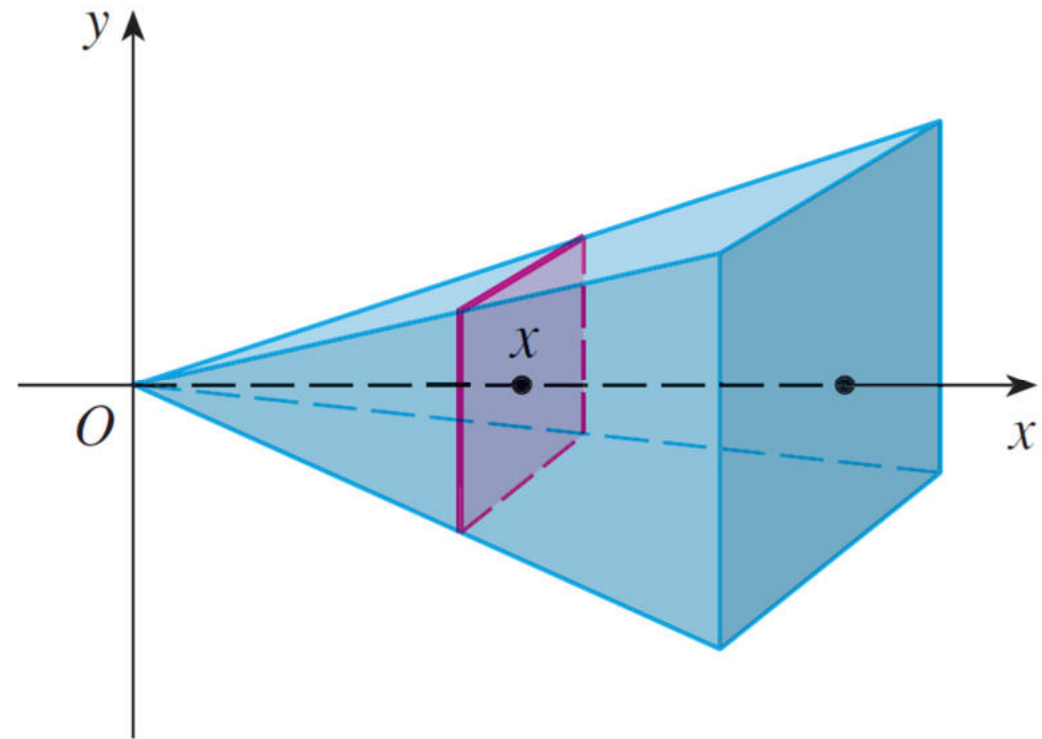
Slicing Method



The Method of Slicing – on X axis

- ▶ The volume of a solid
- ▶ from $x = a$ to $x = b$
- ▶ with a cross-sectional area $A(x)$ is

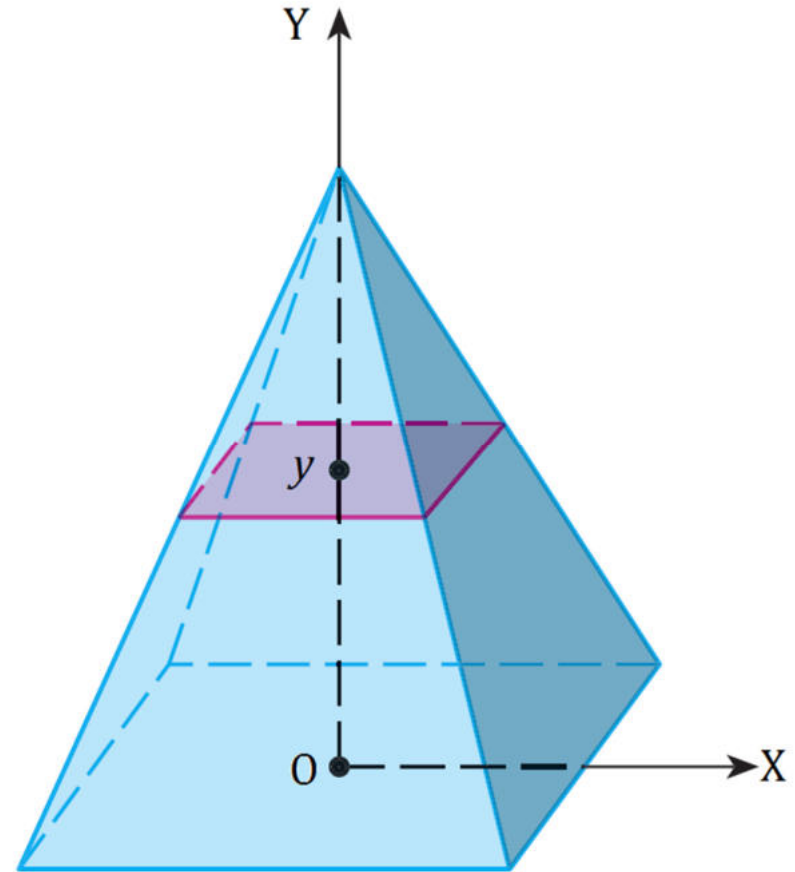
$$V = \int_a^b A(x) \, dx.$$



The Method of Slicing – on Y axis

- ▶ The volume of a solid
- ▶ from $y = a$ to $y = b$
- ▶ with a cross-sectional area $A(y)$ is

$$V = \int_a^b A(y) \, dy.$$



Procedure for the method of Slicing

► If we want to find the Volume of a Solid using Slicing method, we will follow following steps:

Sketch the solid with a typical cross-section

Find the **area** of a cross-section i.e. $A(x)$ or $A(y)$

Find the **limits** of integration i.e. " a " & " b "

Integrate $A(x)$ or $A(y)$ to find the volume.

Method 11 $\Rightarrow$ Example – 2 2. Find the area of a cross-section i.e. $A(x)$ or $A(y)$.

Using slicing method determine volume of a sphere of radius "r".



Solution:

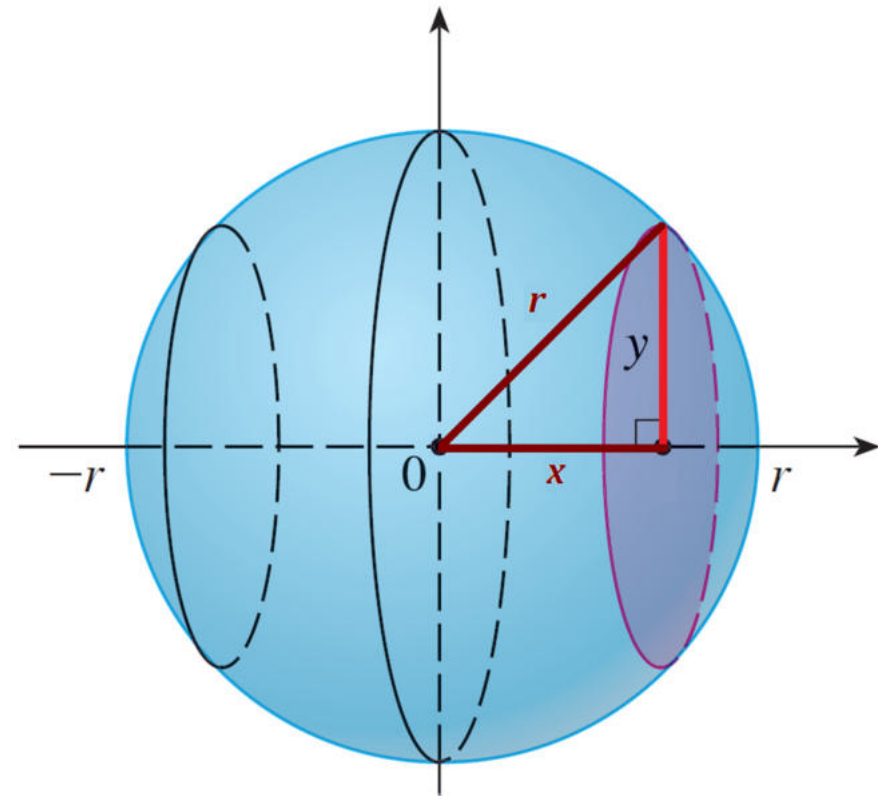
Using the Pythagorean Theorem,

The radius of cross sectioned circle is

$$\text{Radius} = y = \sqrt{r^2 - x^2}$$

$$\therefore A(x) = \pi y^2$$

$$\begin{aligned}\therefore A(x) &= \pi \left(\sqrt{r^2 - x^2} \right)^2 \\ &= \pi (r^2 - x^2)\end{aligned}$$



Method 11 $\Rightarrow$ Example – 2 (continue)

4. Integrate $A(x)$ to find the volume.

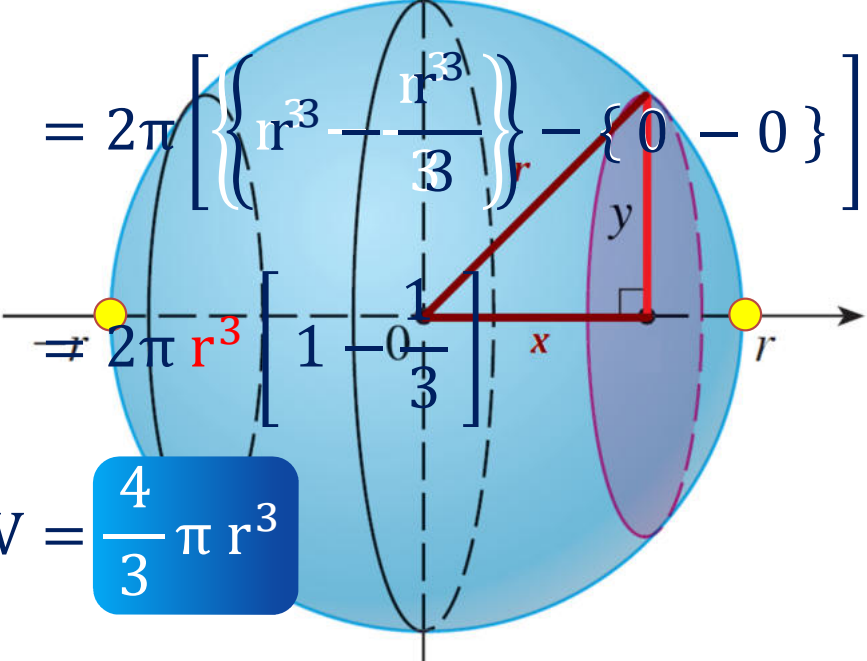
From the figure,

$$a = -r \quad \text{and} \quad b = +r,$$

$$\text{Volume of sphere} = \int_a^b A(x) \, dx$$

$$= \int_{-r}^{+r} \pi (r^2 - x^2) \, dx$$

$$= 2 \int_0^r \pi (r^2 - x^2) \, dx$$


$$V = 2\pi \left[r^2 \cdot x - \frac{x^3}{3} \right]_{x=0}^{x=r}$$
$$= 2\pi \left[\left\{ r^3 - \frac{r^3}{3} \right\} - \{ 0 - 0 \} \right]$$
$$V = \frac{4}{3} \pi r^3$$

Method 11 $\Rightarrow$ Example – 3 2. Find the area of a cross-section i.e. $A(x)$ or $A(y)$.

Derive the formula for the volume of a right pyramid whose altitude is “ h ” and whose base is a square with sides of length “ a ”.

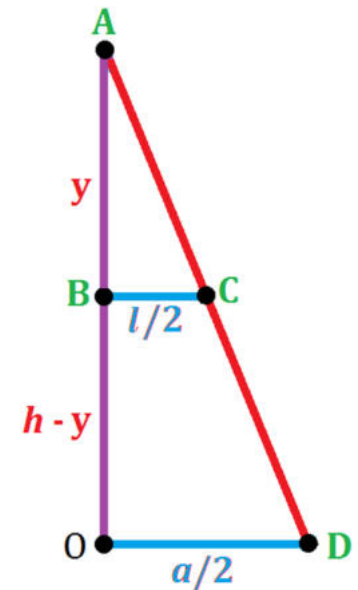


Solution:

Using property of similar triangles,

$$\frac{BC}{OD} = \frac{AB}{AO} \Rightarrow \frac{\cancel{\frac{l}{2}}}{\cancel{\frac{a}{2}}} = \frac{y}{h} \Rightarrow l = \frac{a \cdot y}{h}$$

$$\therefore A(y) = l^2 = \frac{a^2}{h^2} \cdot y^2$$



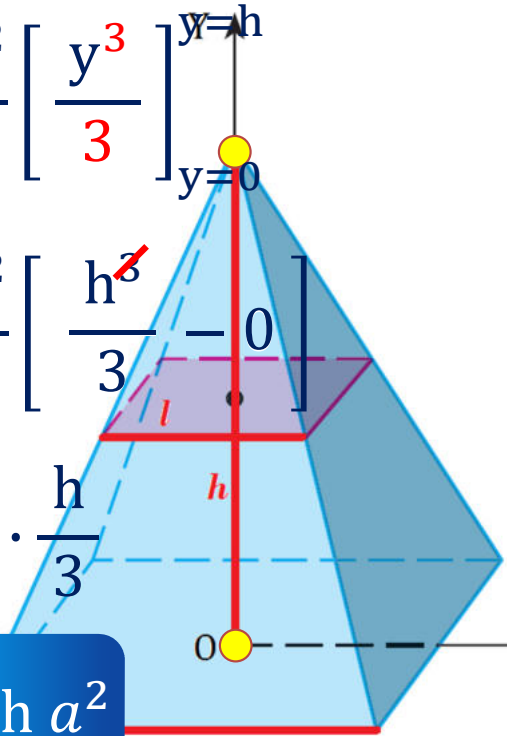
Method 11 $\Rightarrow$ Example – 3 (continue)

4. Integrate $A(y)$ to find the volume.

From the figure,

$$a = 0 \quad \text{and} \quad b = h,$$

$$\begin{aligned} \text{Volume of pyramid} &= \int_a^b A(y) \, dy \\ &= \int_0^h \frac{a^2}{h^2} \cdot y^2 \, dy \\ &= \frac{a^2}{h^2} \int_0^h y^2 \, dy \end{aligned}$$


$$\begin{aligned} V &= \frac{a^2}{h^2} \left[\frac{y^3}{3} \right]_{y=0}^{y=h} \\ &= \frac{a^2}{h^2} \left[\frac{h^3}{3} - 0 \right] \\ &= a^2 \cdot \frac{h}{3} \\ V &= \frac{1}{3} h a^2 \end{aligned}$$

Volume of Solid by **Disk Method**

